Hemant Rajpal

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Software Engineer and Founder of Perseverance AI, specializing in high-performance computing (HPC), AI-driven drug discovery, and scalable cloud-based AI solutions. With 11+ years of experience in machine learning, distributed computing, and MLOps, I have developed multi-cloud AI architectures, optimized GPU-intensive AI pipelines, and automated enterprise-scale ML workflows. I am proficient in Python, TensorFlow, PyTorch, and cloud platforms (AWS, Azure, GCP), I focus on building AI models for drug discovery, predictive analytics, and real-time AI inference.
As the Founder of Perseverance AI, I lead research in AI-powered molecular screening, computational drug design, and automated high-throughput simulations to accelerate novel therapeutic discovery. I am passionate about bridging AI research with real-world applications, and have a strong interest in expanding into robotics, autonomous systems, and generative AI for intelligent drug discovery.

Links and GitHub

- GitHub: github.com/HemantRajpal-9018
- High-throughput Molecular Docking Pipeline (HPC, Gaussian, AutoDock Vina): github.com/HemantRajpal-9018/drug_discovery | Video: youtu.be/nmgoLEuEIH4
- Full Stack Chemical Structure Search Platform (React, TypeScript, Supabase, Ketcher): repo will be added when public
- Heart Attack Prediction (Python, scikit-learn, XGBoost, FastAPI, Docker): github.com/HemantRajpal-9018/Heart-Attack-Prediction

Experience

2024 – PRESENT

AI Software Engineer | Synovellab LLC | New Haven. Connecticut

- Architected and implemented AI and ML pipelines that accelerated ligand identification by ~35 percent.
- Developed GNN architectures for protein ligand interaction prediction, improving accuracy over baselines.
- Used distributed PyTorch with NCCL on AWS GPU clusters to reduce training time by ~50 percent; exported models for optimized inference.
- Productionized workflows with containers and Kubernetes; added CI/CD and monitoring for reliable retraining and releases
- Deployed and managed machine learning models using **Azure Machine Learning (AML)** workspace with versioned pipelines and automated retraining.
- Utilized **Azure Data Lake Storage** and **Azure Blob Storage** for scalable storage of large training datasets and model artifacts.
- Leveraged **Azure Synapse Analytics** and **Azure Databricks** for distributed training and data preprocessing.
- Automated data ingestion from various sources using **Azure Data Factory (ADF)** pipelines integrated with ML workflows.
- Designed and maintained CI/CD pipelines using **Azure DevOps Pipelines** for end-to-end ML lifecycle automation—from data ingestion to deployment.
- Integrated **GitHub Actions / Azure Repos** with ML workflows for version control of models, experiments, and datasets.
- Prototyped a protein ligand RAG assistant using sentence transformer embeddings and vector search with LangChain or LlamaIndex to ground answers in literature and assay notes.
- Introduced MLflow for experiment tracking and DVC for data and asset versioning to improve reproducibility and rollback safety.

MARCH 2023 – FEBURARY 2024

Cloud Engineer | Cloudflare | Washington D.C, United States

- Engineered and deployed scalable high-performance computing frameworks utilizing Kubernetes and Terraform, significantly improving resource efficiency in distributed computational environments.
- Created robust CI/CD pipelines employing Jenkins and GitLab CI, facilitating automated deployments and drastically minimizing service downtime.
- Implemented proactive system monitoring and reliability enhancements using Prometheus and Grafana.
- Built advanced real-time anomaly detection infrastructures employing Kafka, Spark Streaming, and Elasticsearch, significantly augmenting fraud detection capabilities.
- Deployed RESTful endpoints for ML models using **Azure Kubernetes Service (AKS)** and **Azure Functions** for real-time inference.
- Implemented model monitoring and drift detection using **Azure Monitor** and **Application Insights**.
- Secured ML workflows using **Azure Key Vault**, **Managed Identities**, and role-based access control (RBAC).
- Automated testing, packaging, and deployment of ML models using **Docker**, **Kubernetes**, and **Helm** within Azure environments.
- Employed **Terraform / Bicep** for Infrastructure as Code (IaC) to provision ML environments on Azure.
- Used **MLflow** integrated with Azure DevOps for experiment tracking, model registry, and reproducibility.
- Implemented automated model retraining and validation pipelines triggered by new data availability or performance drift.
- Conducted A/B testing and canary deployments of ML models using **Azure Traffic Manager** and **AKS**.

JULY 2021 – AUGUST 2022

Graduate Research Assistant | University of New Haven | New Haven, United States

- Designed and implemented end-to-end machine learning pipelines using Python to support predictive modeling tasks in scientific research environments.
- Applied advanced AI techniques including unsupervised clustering, dimensionality reduction (PCA), and neural networks to identify patterns in complex scientific datasets and accelerate decision-making in experimental workflows.
- Developed and deployed deep learning models for pattern recognition in quantum simulation data, improving model accuracy and interpretability in high-dimensional spaces.
- Managed a 128-node high-performance computing (HPC) cluster using SLURM for distributed AI workloads, optimizing resource allocation and parallel training of machine learning models.
- Built full-stack AI-enabled web applications with Django and RESTful APIs, integrating real-time data visualization and secure user access to support collaborative AI research initiatives.
- Contributed to two peer-reviewed publications combining AI methodologies with domain-specific scientific research, demonstrating the application of machine learning in interdisciplinary environment.
- Built scalable ML pipelines using **Azure ML Pipelines**, automating data preprocessing, training, validation, and deployment stages.
- Optimized distributed training of deep learning models using **Azure GPU-enabled VMs** and **Azure Batch AI**.
- Enabled seamless collaboration between data scientists and engineers using **Azure Notebooks**, **AML Compute Instances**, and **AML Datasets**.
- Managed model versioning and lineage using **Azure ML Model Registry**, ensuring reproducibility and traceability of ML experiments.

MARCH 2018 – DECEMBER 2020

DevOps Engineer | Google | Gurugram, India

- Directed complex cloud migrations to AWS and Azure, substantially modernizing infrastructure and enhancing operational efficiencies.
- Oversaw containerization strategies using Docker and Kubernetes, resulting in 40% faster deployment turnaround.
- Standardized Infrastructure as Code (IaC) deployments through Terraform and Azure Resource Manager (ARM), ensuring uniformity and consistency across diverse environments.
- Developed fully automated CI/CD pipelines utilizing Jenkins and GitLab, significantly reducing manual interventions and increasing operational productivity.
- Created Python-based automation scripts to streamline deployment processes, configuration management, and infrastructure provisioning.
- Engineered Python-driven analytics tools for proactive monitoring, anomaly detection, and system performance optimization.

AUGUST 2014 – FEBRUARY 2018

Software Engineer | Bharti Infratel Ltd | Gurugram, India

- Designed and implemented scalable enterprise web applications leveraging Java, Spring Framework, and Angular, improving overall system performance and user engagement.
- Integrated comprehensive real-time monitoring solutions with ELK Stack, Prometheus, and Grafana, facilitating advanced insights into network and system performance metrics.
- Developed RESTful APIs using Spring Boot, significantly enhancing microservices interoperability and scalability.

Projects

High-Throughput Molecular Docking Platform

- End to end workflow from SMILES to pose scoring with Open Babel, Gaussian, and AutoDock Vina on SLURM or MPI.
- Parallel Gaussian jobs and batched Vina runs improved throughput on multi target screens; reproducible configs and logs.
- Utilities for ligand preparation, receptor prep, scoring aggregation, and error handling; ready to containerize and track with MLflow.
- Optional API layer with FastAPI to submit ligands and retrieve scores and pose files; simple autoscaling on Kubernetes.
- Repo: github.com/HemantRajpal-9018/drug_discovery | Video: youtu.be/nmgoLEuEIH4

Full Stack Chemical Structure Search Platform (React, TypeScript, Supabase, Ketcher)

- Chemistry native UX using Ketcher for draw and edit; SMILES or Molfile queries across PDFs and public sources like PDB.
- PDF ingestion and entity extraction with PDF.js; RAG style retrieval via LlamaIndex; GPT 4o Vision assistant to explain matched structures and figures.
- Service Manager abstraction to switch between mock and real backends; hardened with Supabase auth and telemetry events.
- Reusable TypeScript modules for structure standardization and similarity estimation; planned RDKit and vector database integration.
- Repo: will add when public

Heart Attack Prediction (Python, scikit learn, XGBoost, FastAPI, Docker)

- Built an end to end tabular ML pipeline for risk prediction: data cleaning, feature engineering, split strategy, and model selection across Logistic Regression, Random Forest, and XGBoost.
- Addressed class imbalance with stratified splits and threshold tuning; evaluated with ROC AUC, F1, precision, and recall; Best model achieved ROC-AUC 0.87 and F1 0.72 on a held-out test set
- Logged experiments to MLflow with parameters, metrics, and artifacts; versioned datasets and models for reproducibility.

- Exposed a FastAPI inference service with schema validation and input checks; containerized with Docker for local or cloud deploy.
- Added simple Streamlit UI and SHAP explanations to improve transparency for non technical users; planned CI and Prometheus metrics.
- Repo: github.com/HemantRajpal-9018/Heart-Attack-Prediction

Multi Cloud AI Deployment Platform

- Engineered a multi-cloud AI deployment architecture utilizing Azure DevOps, AWS Lambda, and Docker Swarm, achieving high availability (99.9%) and robust scalability for diverse enterprise applications.

Autonomous Monitoring Framework

- Developed and implemented an innovative hybrid monitoring platform combining Prometheus and custom Python-based analytics, significantly reducing downtime and enabling proactive system performance optimization by approximately 30%.

Skills

- Languages: Python, SQL, Bash
- ML and DL: PyTorch, TensorFlow, scikit-learn, XGBoost, CNNs, RNNs, GNNs (GCN, GAT, MPNN)
- GenAI: LLMs, RAG, embeddings, LangChain, LlamaIndex, vector search (FAISS, pgvector), prompt patterns, LoRA fine tuning
- Serving and performance: ONNX Runtime, TensorRT, quantization, A/B testing, Dockerized inference on Kubernetes
- Data and pipelines: Spark, Kafka, Airflow, Pandas, Delta workflows, ETL and feature engineering
- MLOps: MLflow, DVC, model registry, CI/CD with GitHub Actions and Jenkins, monitoring with Prometheus and Grafana
- Cloud and infra: AWS, GCP, Azure, Docker, Kubernetes, Terraform, cost and reliability optimization
- Cheminformatics: Open Babel, AutoDock Vina, Gaussian, SMILES, MOL/SDF, PDB, PDBQT, pose scoring and docking workflows

Education

- DECEMBER 2022
  Master of Science, Computer Science| University of New Haven | New Haven, CT
  3.8 GPA | Dean Scholarship 75%
- JUNE 2014
  Bachelor of Technology, Computer Science |Kurukshetra University | Kurukshetra, India
  3.6 GPA

Certifications

- AWS Certified Machine Learning – Specialty, AWS (In progress)
- AWS Certified Solutions Architect – Associate, AWS (Achieved 01/2025)
- Microsoft Certified: Azure Developer Associate, Microsoft (Achieved 12/2023)
- Neural Networks and Deep Learning, DeepLearning.AI
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization, and Optimization, DeepLearning.AI
- Structuring Machine Learning Projects, DeepLearning.AI
- Sequence Models, DeepLearning.AI
- Convolutional Neural Networks, DeepLearning.AI

Accomplishments

- Co-inventor on a U.S. Provisional Patent (63/685,817) filed with UNOGEN BIOTECH LTD., titled "Methods and Systems for Engineering Kinase Inhibitors," introducing a novel computational workflow for large- scale docking simulations to accelerate drug discovery
- High-Throughput Molecular Docking Platform: Developed a scalable molecular docking framework integrating MPI, SLURM, and Python, enabling rapid multi-kinase inhibitor discovery.